SEC P vs NP log 2026-04-29

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-29 01:46 UTC

SEC P vs NP — daily log 2026-04-29

5 cycles recorded

GF(2)-Rank Defect of Term-Indicator Matrix Lower-Bounds Monotone DNF for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid theory — the GF(2) vector matroid M_F realized by the term-indicator matrix $A_F \in F_2^{m\times N}$ of a monotone DNF $F = \vee$*{i=1}^m T_i (rows* □$\{T_i\}$, columns indexed by variables). Its rank $\rho_2(F) := \mathrm{rank}_{F_2}(A_F)$ is the canonical computable matroid invariant in the Whitney–Welsh tradition; the corresponding 'rank defect' $\Delta(F) := \log_2(m(F)+1) - \log_2(\rho_2(F)+1)$ is a polymatroid-style quantity (subadditive log-rank vs. log-cardinality) that is rarely deployed in monotone circuit lower bounds (Razborov 1985 used Bollobás set-pair / sunflower closure, not GF(2) span of clutter indicators). × Monotone DNF representation size of the k-CLIQUE_n indicator on $N = C(n,2)$ edge variables; equivalently, the minimum-cardinality clutter of minimal terms whose union gives k-CLIQUE_n.
- **Recorded**: 2026-04-29 00:11 UTC
- **Entry ID**: `023dd98c00ee`

Statement

Let F be a monotone DNF on N variables with minimal-term family $H_F = \{T_1,\ldots,T_m\}$, A_F its $m\times N$ indicator matrix, and define $\Delta(F) := \log_2(m+1) - \log_2(\mathrm{rank}_{F_2}(A_F)+1)$. Conjecture: (a) $\Delta(F\wedge G) \le \Delta(F) + \Delta(G) + 1$ and $\Delta(F\vee G) \le \max(\Delta(F),\Delta(G)) + 1$ (approximate submodularity through Boolean operations); (b) $\Delta(F) \le \log_2(N+1)$ for any monotone DNF with $m \le \mathrm{poly}(N)$ (since $\mathrm{rank}_{F_2} \ge 1$ and $m \le \mathrm{poly}$); (c) for the canonical k-CLIQUE_n DNF $F^{n,k}$ with $k = \lfloor n/2\rfloor$, $\Delta(F^{n,k}) \ge n/2 - 3\cdot\log_2 n - 4$ for all $n \ge 6$, witnessing a $\Omega(n)$ gap between log-cardinality and log-GF(2)-rank.

Rationale

The GF(2) span of clique-edge-indicator vectors collapses to a sub-quadratic-rank object (rank $\le C(n,2)$), while the minimal-term count $C(n,\lfloor n/2\rfloor) \approx 2^n/\sqrt{n}$ is exponential — so the LOG ratio is $\Theta(n)$, much larger than any rank deficit attainable by poly-size clutters. Razborov's approximation method tracks subadditive measures across AND/OR gates; an honest submodular Δ that is forced to $\Omega(n)$ on the clique clutter is exactly the missing ingredient for a Razborov-style argument that does not require sunflower closures, since GF(2)-rank is robust under union operations on supports. The bridge from matroid representability to monotone DNF size is concrete (one Gaussian elimination per gate), avoiding the natural-proofs / algebrization barriers that block spectral or Fourier approaches.

Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2210.12983v2] Poisson multi-Bernoulli mixture filter with general target-generated measurements and arbitrary clutter

Empirical Test

- exit code: 1, elapsed: 0.10s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 118, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 78, in run_trial
   F_random = random_poly_dnf(n, n, 2) + random_poly_dnf(n, n**2, 3) + random_poly_dnf(n, n * math.flo
              ^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b7aab2a.py", line 56, in random_poly_d
    selected_edges = random.sample(edge_set, w)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

The test crashed with a TypeError in random_poly_dnf (random.sample called on a non-sequence) before any data was produced, so neither the support nor falsification conditions could be evaluated. | next: Fix random_poly_dnf by passing a list (e.g., list(edge_set) or sorted(edge_set)) to random.sample and rerun the 30-seed sweep.

Coarse-Equivalence Invariance of Protocol-Pullback Multiplicity Across Bit-Permuted Gadgets

- **Verdict**: INCONCLUSIVE
- **Bridge**: Coarse Geometric Lifting (CGL) — Coarse Geometry / Roe Algebras (within Geometric Group Theory) × Deterministic 2-party communication complexity, specifically the multiplicity m_Π of the RoeCover extracted from a deterministic protocol's ProtocolPullback for the lifted function f∘G^n
- **Recorded**: 2026-04-29 00:24 UTC
- **Entry ID**: aa9fe4edbfe5

Statement

Let G_1 = (X, Y, g, d) be a MetricGadget on small alphabet (|X|,|Y| ≤ 4) and let G_2 be obtained from G_1 by a bi-Lipschitz bijection φ: X×Y → X×Y with distortion ≤ 2 (a coarse equivalence: φ permutes inputs while distorting d by at most a factor 2). Fix any boolean f: {0,1}^n → {0,1} with n ≤ 4. Then for every deterministic protocol Π_1 computing f∘G_1^n, there exists a deterministic protocol Π_2 computing f∘G_2^n with cost cost(Π_2) ≤ cost(Π_1) + O(n), and the multiplicities of their CoarsePullback covers at scale R = diam(G) satisfy m_{Π_2} ≤ 2 · m_{Π_1} + 1. Equivalently, CLC(f,G_1) and CLC(f,G_2) agree up to a constant factor independent of f, instantiating Axiom A1

(Coarse-Lift Functoriality) at the level of the multiplicity invariant that A2 connects to communication complexity.

Rationale

This conjecture directly tests Axiom A1 (Coarse-Lift Functoriality) by asserting that the *multiplicity invariant* m_Π exposed by Axiom A2 (Protocol → Cover) is preserved under coarse equivalences of the underlying gadget. If A1 holds, a bi-Lipschitz φ on X×Y lifts coordinatewise to a bi-Lipschitz map on (X×Y)^n with the d_⊕ metric (distortion preserved since ▯^{1} products of bi-Lipschitz maps remain bi-Lipschitz), so any Roe-style cover at scale R for the G_1-lift can be transported to a cover at scale ≤ 2R for the G_2-lift while at most doubling the multiplicity (each transported set may merge with one neighbor at the boundary, contributing the +1 slack). The O(n) cost overhead in Π_2 reflects simulating φ coordinatewise. Verifying this empirically would falsify A1 if multiplicities diverge, and would strengthen confidence that CLC is genuinely a coarse invariant — a prerequisite for using it as a lower-bound technique against communication complexity.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2408.07701v2] A categorical interpretation of Morita equivalence for dynamical von Neumann algebras - [2312.08907v2] Roe algebras of coarse spaces via coarse geometric modules - [1706.00276v1] Counting coarse subsets of a countable group - [2110.02008v2] Lifted Reed-Solomon Codes and Lifted Multiplicity Codes - [1903.06084v8] Lifting coarse homotopies

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 49, in run_trial
    phi_perms = generate_bi_lipschitz_permutations()
                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 35, in generate_bi_l
    if is_bi_lipschitz(phi):
       ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 24, in is_bi_lipschi
    if dist(phi[(x1, y1)], phi[(x2, y2)]) > distortion * dist((x1, y1), (x2, y2)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8a94fc38.py", line 18, in dist
    return max(abs(ai - bi) for ai, bi in zip(a, b))
                                          ^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in the bi-Lipschitz check (dist() called on integers instead of tuples) before any trials produced data, so neither the support nor falsification criteria can be evaluated. | next: Fix the dist() helper to handle the (x,y) pair representation correctly (ensure phi maps tuples to tuples and dist unpacks both arguments), then rerun the 5-seed sweep.

Persistent H_0 Component Count of Sign-Quotient Row Set Caps Real Rank

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Persistent homology / Vietoris-Rips topology of finite metric spaces (Vietoris 1927; Edelsbrunner-Letscher-Zomorodian 2002; Carlsson-Zomorodian 2005; Carlsson 2009). The 0-th persistent Betti number β_0(VR(X,ε)) of a finite metric space (X,d) at scale ε is the number of connected components of the ε-thresholded distance graph, computable by Kruskal/union-find in O(|X|^2 α(|X|)). Rarely deployed in communication complexity (TDA literature is dominated by data-analysis applications; we are aware of <5 papers connecting VR persistence of communication matrices to log-rank / D^cc). × Real rank rk_R(M_f) of the 2^k × 2^k ±1 communication matrix M_f[x,y]=(-1)^{f(x,y)} of a Boolean function f:{0,1}^k × {0,1}^k → {0,1} — the canonical Lovász log-rank / Mehlhorn-Schmidt target for deterministic two-party communication complexity D^cc(f) ≥ log_2 rk_R(M_f).
- **Recorded**: 2026-04-29 00:40 UTC
- **Entry ID**: `bc750201aaf2`

Statement

Let X_f^± ⊂ {±1}$^{\{2}$k}/{±1} be the set of distinct rows of M_f modulo global sign, equipped with the projective Hamming distance d_±(r,s) = min(d_H(r,s), 2^k - d_H(r,s)). Let τ_±(M_f) := β_0(VR(X_f^±, 2^{k-2})), the number of components of the graph linking r,s whenever d_±(r,s) ≤ 2^{k-2}. Then for every Boolean f and every k ∈ {3,4,5,6}, τ_±(M_f) ≤ rk_R(M_f); a single (k,f) with τ_±(M_f) > rk_R(M_f) refutes.

Rationale

Rank-r ±1 matrices have all rows lying in an r-dimensional subspace of R^N, and ±1 vectors in such a subspace are constrained to a structured (centrally-symmetric) finite set whose pairwise distances avoid the 'mid-Hamming' regime around N/2 only by sitting in a small affine pattern; persistence at scale N/4 should therefore not be able to manufacture more than r distinct components after the ±-quotient. This bridges the topological 'shape-counting' of the row metric space to the algebraic rank-bound side of log-rank, a structural inequality complementary to spectral approaches and untouched by natural-proofs (rk_R is the property being lower-bounded, not a hardness predicate on truth tables) and untouched by algebrization (no oracle relativization is used).

Novelty

- Judge: `NOVEL` over 5 arXiv hits

Top hits consulted: - [LIPIcs.ESA.2024.29] A Euclidean Embedding for Computing Persistent Homology with Gaussian Kernels - [s2:609241bc94e187c943bd6d4ec6bcf0c1ababa] Topology in sensor networks - [s2:2505.16879] How high is 'high'? Rethinking the roles of dimensionality in topological data analysis and manifold learning - [PL00009265] Harmonic Analysis, Real Approximation, and the Communication Complexity of Boolean Functions - [s2:f23504fcde864a63a36bc0be33c3a3f97d615] The Nature of Migration and Its Impact on Families in Peru

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

 File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7aad0d36.py", line 163, in run_trial
    M.append([Fraction(row[i]) for i in range(n)])
```

```
                        ~~~^^^
IndexError: tuple index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any RESULT line, so neither the support nor the falsification condition could be evaluated. Per rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the row-construction bug (tuple index out of range when building M from row entries) and re-run the full 4×30×6 grid to obtain τ_± and rk_R counts.

Secant Rank Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic geometry of secant varieties × Communication complexity
- **Recorded**: 2026-04-29 01:14 UTC
- **Entry ID**: `1a21dd39fb6f`

Statement

The secant rank of the communication matrix for the Disjointness problem DISJ_n is Ω(n), implying a Ω(n) lower bound on randomized communication complexity.

Rationale

Secant rank measures the minimal number of rank-1 tensors needed to approximate a matrix, directly linking to communication complexity via the log-rank conjecture. For DISJ_n, its matrix structure forces high secant rank due to combinatorial constraints, providing a novel algebraic invariant for proving lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.82s`

```
me': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'conjecture_holds': Tr
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8374.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.5, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.833333333334, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8373.666666666666, 'instances_tested': 6, 'con
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8374.0, 'instances_tested': 6, 'conjecture_hol
TRIAL: {'metric_name': 'secant_rank', 'metric_value': 8372.833333333334, 'instances_tested': 6, 'con
RESULT: SUPPORTED mean=8373.511111111111 std=0 support_fraction=1.0
```

Judge reasoning

The test used n=6, which is too small to establish Ω(n) bounds. Metric values show no scaling with n, suggesting potential saturation or hidden structure in small cases. | next: Test with larger n (e.g., n=128) to observe scaling behavior of secant rank

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Sign-Rank Distinguishing Tensor for Read-Twice Branching Programs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: discrepancy theory × matrix rigidity
- **Recorded**: 2026-04-29 01:46 UTC
- **Entry ID**: `d2ba6b1a27c2`

Statement

For any read-twice branching program P computing a Boolean function f, the sign-rank of its communication matrix M_P is O(log size(P)), but for the inner product mod 2 function IP_2, the sign-rank of its trivial read-twice BP is Ω(n).

Rationale

This conjecture leverages the connection between sign-rank (a discrepancy-theoretic invariant) and matrix rigidity (a field_B concept). The IP_2 function's exponential lower bound for read-once BPs suggests its read-twice representation must exhibit high sign-rank, while read-twice BPs for general functions should have logarithmic sign-rank due to their structured computation. This bridges the gap between BP complexity and communication complexity barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 100, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 77, in run_trial
    A = generate_random_sign_matrix(n, rank)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 47, in generate_rand
    U = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2cd38001.py", line 30, in gaussian_elim
    factor = Fraction(A[j][i], A[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(0, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError during Gaussian elimination, preventing data collection | next: Debug gaussian_elimination() to handle zero pivots and regenerate matrices